

Wally: Fine-Tuning a Sentence Transformer with Wavelength to Assess Linguistic Semantic Reasoning

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ABSTRACT

Sentence transformers, derived from the ever-evolving Large Language Model (LLM) architecture, still possess limitations in capturing human intuition and vocabulary semantics. To assess the limitations of simulating human cognition, I created Wally, a fine-tuned sentence embedder trained on the mechanics of the trending party game, *Wavelength*. By unfreezing succeeding layers, appending a regression output node, and calibrating semantic similarities, Wally showcases a strong grasp on replicating human behavior. Wally’s best implementation boasts a 53.41% MAE and 92.11% performance increase compared to baseline models. Wally’s training, evaluation, and experiments are openly available online. Additionally, a fully-interactive Flask website was deployed, allowing users to play *Wavelength* games with Wally.

1. INTRODUCTION

Sentence transformers in tandem with LLMs have seen a widescale domination across the technological realm. By converting sentences into a fixed-size vectors and applying a neural network architecture, computers gained the ability to naturally interpret text like a human. Conversely, however, there are areas in linguistics that these architectures fail to properly comprehend such as: pragmatic reasoning, semantic similarities, literary devices, and contextual language [1].

To visualize and analyze potential linguistic performance, I built a model, Wally, revolving around the popular verbal communication game, *Wavelength*. In the game, one player (guesser), tries to guess a hidden numerical value among a range via a dial. The guesser is given a card of two opposing concepts (e.g., "hot / cold", "funny / boring", etc.). Additionally, the second player (the psychic) is aware of the hidden value, and must give a verbal clue to implicitly sway the guesser in selecting the correct value (e.g., "lava", "clown", etc.). After contemplating the strength of semantic similarity between their clue and given opposing concepts, the guesser will finally turn their dial needle where they believe the correct value is. The value is then unveiled, and the guesser is

awarded points based on how close their needle was. Players alternate turns being the guesser until one of them acquires an agreed-upon, arbitrary number of points [3].

Given the nature of this game, players must be able to process complex reasoning within linguistics and human behavior. As current applications of sentence transformers often fail in these environments, *Wavelength*’s structure serves as an appropriate training ground for human behavior replication.

1.1. Sample Game

Guesser: I will draw a word-pair card. I got "Hot/Cold".

Psychic: I will peek, such that I know the hidden value. It’s relatively close to the left. As such, I will tell the clue "lamp", since it’s a little hot.

Guesser: I deduce that the dial must be slightly closer to the left, since I think "lamp" is kind of hot. I turn the dial and unveil the value. It is on the exact range. I get maximum points for this turn.

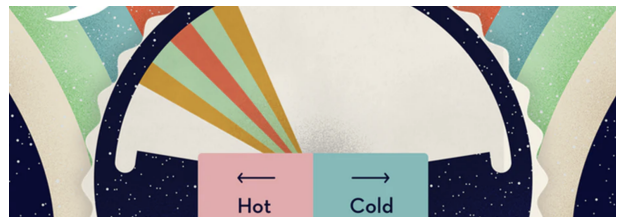


Fig. 1. The psychic must give a clue that would allow the guesser to deduce the correct value. This value is hidden from the guesser until after their guess.

2. DATA AUGMENTATION

2.1. WavelengthEval Dataset

The main inspiration for the data behind Wally’s training is *WavelengthEval* [2]. Each row is composed of an left anchor word, a right anchor word, a clue, a target value, list of participant answers as a guesser, and the response mean. In total, there are two clues for each of the fifty word pairs, totalling 100 rows of data.

¹ Code available at <https://github.com/Akigaii/wally>.

Table 1. Wavelength-Eval General Information

Property	Value
Total Entries	100
# of Word Pairs	50
# of Clues / Word Pair	2
# of Responses / Row	40

2.2. WavelengthEval Issues

As *WavelengthEval* was intended to assess current LLMs, there are several detriments that would make it unfeasible for training Wally:

- The overall dataset size is small, only 100 rows.
- Some word pairs are more subjective/objective (i.e. "Blue/Green" vs. "Boring/Exciting").
- Some word pairs are very similar (i.e. "Bad/Good" and "Bad for you/Good for you").
- Proper nouns and official names are utilized (i.e. "Big Mac", "Patreon", "Walmart").

2.3. Data Collection

While *WavelengthEval* is an excellent jumping off point, the demands of Wally’s training warrants collecting additional data. As such, before training, I played and documented multiple games of *Wavelength*, applying the following preprocessing steps:

- Cherrypick and generate 30 total word pairs, 15 subjective and 15 objective.
- Generate 50 clues and arbitrary scores for all word pairs.
- Survey such that there are 10 entries for all new clues in `response_list`.
- Average all new `response_list` entries and generate `response_means`.
 - `response_mean` will act as the truth value; the arbitrary score during initialization was only used to gauge a certain area where the predictions might lie.

After capturing all of these *Wavelength* games, the final dataset contains 1500 rows, serving a more robust foundation for the purpose of Wally.

2.4. Flipping Anchor Pairs

A common pitfall of sentence transformers is a tendency for cosine similarity scores to gravitate toward zero when comparing dissimilar pairs. This is because dissimilar word pairs are far more prevalent in the English language than similar ones. To counteract this behavior, the dataset will be doubled, totalling 3000 rows. Afterward half of the rows will have their anchor word pairs switched, and all numerical values will be replaced with their complement, effectively nullifying any imbalance from the dissimilarity bias.

3. METHODS

3.1. Model Architecture

Wally fine-tunes the pretrained `all-mpnet-base-v2` sentence transformer for regression on the *Wavelength* task. It will always serve as the "guesser" and the user will be the "psychic". Wally takes in a selected word pair and sentence/phrase from the user. Then, it will attempt to guess a numerical value [0, 100] to simulate spinning the dial needle.

`all-mpnet-base-v2` takes in a sentence and embeds it within a 768-dimensional vector space. Embedded vectors closer in this space are deemed more semantically related. For Wally’s task, however, this surplus of vectors and equal consideration of every English word introduces an abundance of noise. In *Wavelength*, the main similarity priorities are the anchor words and given clue. To enforce this behavior in Wally, I unfroze the last four layers in the sentence embedder, with a final regression output node.

For training, I utilized AdamW with a $1e-2$ weight decay as the optimizer and Mean Absolute Error (MAE) for the loss function. A 75/10/15 train/val/test split with shuffled loaders are utilized. Hyperparameter selections and tuning will be further explored in Section 4.

3.2. Baselines

Two baseline approaches were analyzed, cosine similarity and axis projection on the raw sentence embeddings produced by `all-mpnet-base-v2`. Both approaches are used on the entire dataset with a 75/10/15 train/val/test split.

3.2.1. Cosine Similarity

Cosine similarity assesses closeness by measures the angle between two embedded vectors, producing a continuous value [-1, 1]. For each clue, the cosine similarity was computed against both the left and right pole embeddings independently. The resulting similarities were then used to derive a predicted position along the spectrum.

$$\text{cos_sim}(\mathbf{c}, \mathbf{p}) = \frac{\mathbf{c} \cdot \mathbf{p}}{\|\mathbf{c}\| \|\mathbf{p}\|} \quad (1)$$

3.2.2. Axis Projection

Axis projections, contrarily, are constructed by subtracting the left pole embedding from the right pole embedding, effectively defining a directional spectrum in embedding space. The clue embedding is then projected onto this axis, and the scalar magnitude of that projection is used as the predicted value. This is, arguably, the most faithful mathematical representation on how *Wavelength* works.

$$\text{proj} = \frac{\mathbf{c} \cdot (\mathbf{r} - \mathbf{l})}{\|\mathbf{r} - \mathbf{l}\|} \quad (2)$$

4. EXPERIMENTS

4.1. Hyperparameter Configurations

During Wally’s training, a total of 12 experiments were conducted, each adjusting hyperparameters such as epochs, batch size, and learning rates. The raw JSON history output and performance graphs per experiment are readily available on the GitHub. All experiments are as follows:

Table 2. Experiment Hyperparameter Configurations

Experiment	Epochs	Batch Size	Learning Rate
1	50	4	1e-5
2	50	4	2e-5
3	50	16	1e-5
4	50	16	2e-5
5	100	4	1e-5
6	100	4	2e-5
7	100	16	1e-5
8	100	16	2e-5
9	200	4	1e-5
10	200	4	2e-5
11	200	16	1e-5
12	200	16	2e-5

4.2. Evaluation Metrics

4.2.1. Mean Absolute Error (MAE)

The primary metrics and loss function for Wally; MAE measures the average magnitude from predicted values to real values. In this case, MAE visualizes numerically how far the dial needle is from the hidden value.

4.2.2. Pearson Correlation Coefficient (r)

Pearson (r) measures the strength and direction of linearity between continuous variables. A value of 1 represents a perfect linear positive relationship, whereas a value of 0 demonstrates no correlation. In this case, it measures how well Wally’s predictions move in the same direction as the true values.

4.2.3. Within-Color Range Accuracy

Wavelength’s scoring depends if the guesser’s spin needle lands within a color range. The smallest and closest range is represented by blue, where if a user guesses within it, they attain +4 points. The middle range is represented by orange, where it corresponds with +3 points. Lastly, the widest range is yellow, where a guesser would get +2 points. (*Note: these ranges are nested, so if a user guesses within the blue range, they would also be inside orange and yellow*). After designing the wheel as an SVG diagram in Adobe Illustrator, the following ranges were extracted:

- **Blue:** ± 3.1 units ($MAE \leq 3.1$), +4 points.

- **Orange:** ± 9.4 units ($MAE \leq 9.4$), +3 points.
- **Yellow:** ± 15.5 units ($MAE \leq 15.5$), +2 points.

5. RESULTS

Table 3. Raw Dataset Results

MAE	Pearson (r)
10.93	0.904

Table 4. Baseline Results

Similarity	Test MAE	Test Pearson (r)
Cosine Similarity	28.57	0.418
Axis Projection	25.86	0.433

Table 5. Experiment Results - MAE & Pearson(r)

Experiment	Test MAE	Test Pearson (r)
1	15.56	0.730
2	23.30	0.485
3	30.11	0.193
4	13.90	0.803
5	13.31	0.805
6	15.65	0.713
7	17.87	0.676
8	13.08	0.788
9	13.31	0.786
10	14.45	0.743
11	14.49	0.757
12	14.86	0.738

Table 6. Experiment Results - Within Color Accuracies

Experiment	Blue	Orange	Yellow
1	24.17%	57.92%	69.58%
2	13.75%	36.67%	52.92%
3	4.58%	21.67%	35.83%
4	22.50%	58.75%	69.58%
5	23.75%	60.00%	76.67%
6	26.67%	60.83%	68.75%
7	19.58%	50.00%	64.17%
8	27.92%	66.25%	74.17%
9	28.75%	65.00%	75.00%
10	27.50%	64.17%	76.25%
11	25.00%	59.17%	73.75%
12	23.75 %	59.17%	72.50%

6. ANALYSIS

6.1. Experiments Comparison

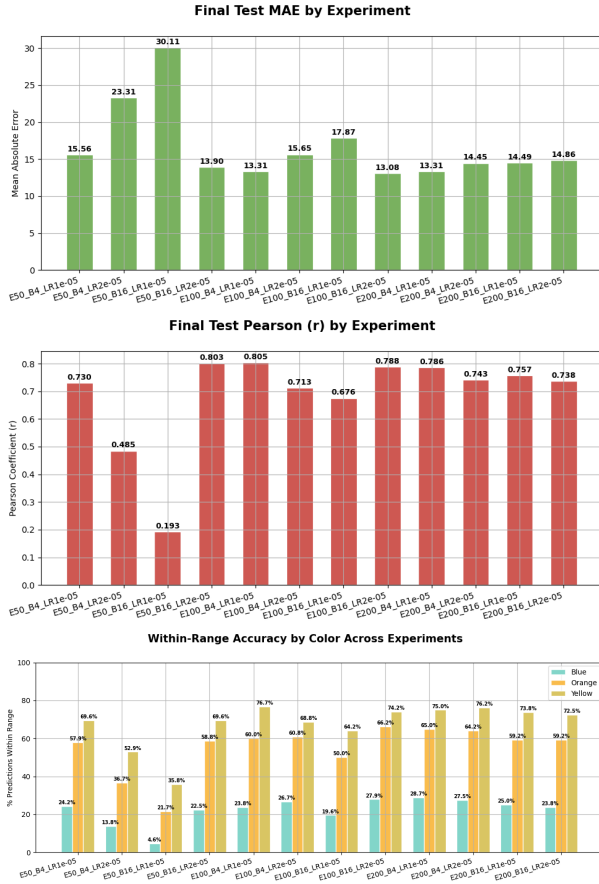


Fig. 2. Experiment comparison across all hyperparameter configurations. (Top) Test MAE, lower is better. (Middle) Test Pearson (r), higher is better. (Bottom) Within-range accuracy by color zone.

6.2. Best Model

Observing the metrics among all experiments, it can be deduced that **Experiment 5** performs the best marginally. It showcases the best balance of low MAE of 13.31, high Pearson (r) of 0.805, and overall high color-range accuracy. As such, this will be the final model used on the Wally website.

Compared to the cosine similarity baselines (MAE: 28.57, Pearson (r): 0.418), Experiment 5 boasts a 53.41% MAE and 92.11% Pearson (r) performance increase, respectively. Compared to the axis projection baselines (MAE: 25.86, Pearson (r): 0.433), Experiment 5 boasts a 48.53% MAE and 85.45% Pearson (r) performance increase, respectively. Given these metrics, it is evident Wally was able to learn semantic similarities and simulate human behavior.

However, Experiment 5's performance still falls short

compared to the raw dataset's metrics (MAE: 10.93, Pearson (r): 0.904). This demonstrates there is still potential for improvement in hyperparameter tweaking. Further elaboration on such will be discussed in Section 6.

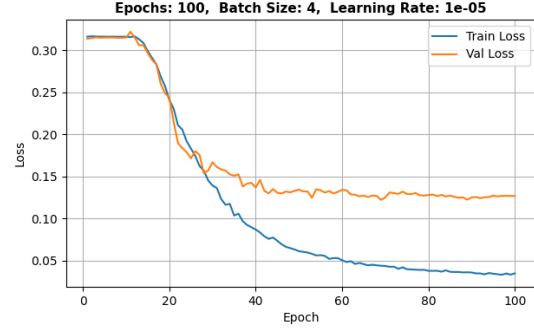


Fig. 3. Train and validation loss curves for Experiment 5. The logarithmic shape and plateauing lines signify a good fit model.

6.3. Worst Model

Conversely, it can be observed that, by far, the worst-performing setup is **Experiment 3**, demonstrating a tremendously high MAE of 30.11, a low Pearson (r) of 0.193, and overall low color-range accuracy.

These results are demonstrably worse than the raw dataset and the baselines.

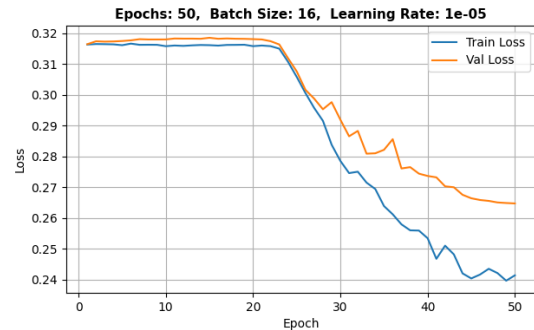


Fig. 4. Train and validation loss curves for Experiment 3. The sharp decline and lack of flattening suggest the model lacked a sufficient warm-up period.

6.4. Hyperparameter Effects

Among the 12 experiments, hyperparameter selection proves to be an imperative decision for the performance of Wally. Assessing the trends shown in Subsection 6.1 makes apparent the following patterns:

6.4.1. Epochs

Epochs had the highest impact of the three hyperparameters. Increasing epochs consistently led to higher performance across all experiments.

6.4.2. Batch Size

Batch size demonstrated a notable effect, where smaller batch sizes (4) generally outperformed larger ones (16).

6.4.3. Learning Rate

Learning rate showed a consistent trend, where $1e-05$ outperformed $2e-05$ across all configurations. This suggests a smaller learning rate is more applicable in sentence transformers.

7. LIMITATIONS

7.1. Subjectiveness

The largest challenge behind training Wally lies within the entire driving nature behind *Wavelength*: opinion. Unlike other regression tasks, *Wavelength* responses inherently have no "correct" answers.

For instance, consider the unused word pair "tastes good/tastes bad" and the clue "burger". Because of the subjective nature of this proposition, depending on the person(s) responding, every possible value $[0, 100]$ becomes a possibility. This ultimately transitions the term "ground truth" to be more of an arbitrary heuristic. The lack of deterministic, objective data leaves Wally unable to capture the various answers of humans simultaneously. Not only is this a handicap for Wally's performance, but it also demonstrates a limitation for human evaluation *Wavelength* itself. This is demonstrated by the raw dataset, showcasing an MAE of 10.93,

The augmented dataset originally had a column to track more objective word pairs (e.g., "messy food/clean food", "low calorie/high calorie", etc.) and subjective word pairs (e.g., "boring/exciting", "bad/good", etc.). While unused for this generalized Wally implementation, further experimentation can potentially unveil stronger performance on objective categories.

7.2. Dataset Scale

Even after augmenting and expanding *WavelengthEval*'s dataset from 100 to 3,000, this size is utterly dwarfed by the infinite combinations of word pairs/clues. Humans have spent every day of their lives developing their vocabulary, connections, and contexts. Starting from scratch, it is impossible for a deep learning model, Wally, to grasp such complex linguistics with only 3,000 terms.

Considering the length it took to acquire these dataset responses, it is apparent that physically documenting games is

unfeasible for further extrapolation. As Wally will be deployed via a website, it is possible to track all input clues to further train Wally at a later date. To further supplement, the Wally gameplay website will soon provide a way for user to insert their own word pairs as well.

7.3. Current Events

Sentence transformers can only process and internalize words at face-value, only considering objective properties like definition. Humans, contrarily, ingest global events and pop culture daily, continuously adjusting their perception of linguistics and human connection.

For instance, consider the word pair "immature/mature". If the user were to input the word clue "67" (the popular internet meme from 2025), an average person would likely understand the nonsensical, "brainrot" history of the term, and turn the needle toward "immature". A deep learning model such as Wally, however, has no way to conceptualize this trend; instead, it only perceives it merely as a number. This further proposes a limitation that a sentence transformer has on replicating human linguistic communication.

8. CONCLUSION

Developing Wally to simulate *Wavelength* shows the promising potential of sentence transformer grasping human linguistics and vocabulary semantics. However, given the incomprehensible performance of human memory, pattern-matching, and real-world contexts, an apparent limitation emerges, begging the question if human nature can truly be replicated by artificial intelligence.

9. REFERENCES

- [1] Linlu Qiu et al., "Wavelengtheval: Benchmarking llms on pragmatic semantic reasoning," *arXiv preprint arXiv:2509.06952*, 2025.
- [2] Linlu Qiu, "Wavelengtheval dataset," <https://github.com/linlu-qiu/wavelengtheval/blob/main/data/data.csv>, 2025.
- [3] Board's Eye View, "Wavelength board game overview," <https://www.boardseyeview.net/post/wavelength>, 2023.